

Problem

Find the solution to the following simultaneous equations:

$$3x_1 - x_2 = 4, \quad -6x_1 + 18x_2 = 16$$

Step-by-step solution

Step 1 of 3

Consider the following set of simultaneous equations,

$$\begin{aligned} 3x_1 - x_2 &= 4 \\ -6x_1 + 18x_2 &= 16 \end{aligned}$$

Arrange the equations in matrix form.

$$\underbrace{\begin{bmatrix} 3 & -1 \\ -6 & 18 \end{bmatrix}}_{\mathbf{A}} \underbrace{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}}_{\mathbf{X}} = \underbrace{\begin{bmatrix} 4 \\ 16 \end{bmatrix}}_{\mathbf{B}}$$

Calculate the determinant of matrix $\mathbf{A}$, Δ .

$$\begin{aligned} \Delta &= \begin{vmatrix} 3 & -1 \\ -6 & 18 \end{vmatrix} \\ &= [3(18)] - [1(6)] \\ &= 54 - 6 \\ &= 48 \end{aligned}$$

Step 2 of 3

Calculate the determinant Δ_1 , which is the determinant of the matrix formed by replacing first column of matrix $\mathbf{A}$ by column matrix $\mathbf{B}$.

$$\begin{aligned} \Delta_1 &= \begin{vmatrix} 4 & -1 \\ 16 & 18 \end{vmatrix} \\ &= [4(18)] + [1(16)] \\ &= 72 + 16 \\ &= 88 \end{aligned}$$

Calculate the determinant Δ_2 , which is the determinant of the matrix formed by replacing second column of matrix $\mathbf{A}$ by column matrix $\mathbf{B}$.

$$\begin{aligned} \Delta_2 &= \begin{vmatrix} 3 & 4 \\ -6 & 16 \end{vmatrix} \\ &= [3(16)] + [4(6)] \\ &= 48 + 24 \\ &= 72 \end{aligned}$$

Step 3 of 3

Calculate x_1 by using Cramer's rule.

$$\begin{aligned}x_1 &= \frac{\Delta_1}{\Delta} \\&= \frac{88}{48} \\&= 1.8333\end{aligned}$$

Calculate x_2 by using Cramer's rule.

$$\begin{aligned}x_2 &= \frac{\Delta_2}{\Delta} \\&= \frac{72}{48} \\&= 1.5\end{aligned}$$

Thus, the solution to the simultaneous equations is $x_1 = 1.833, x_2 = 1.5$.